

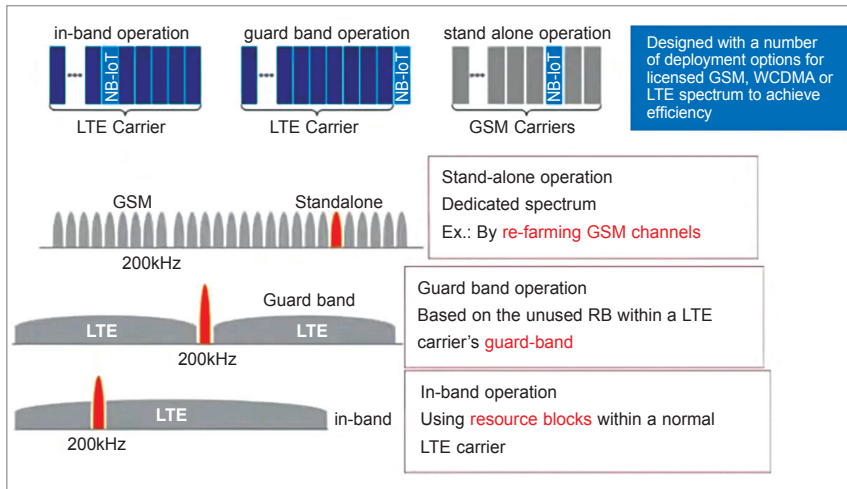


Fig. 5: NB-IoT spectrum and access

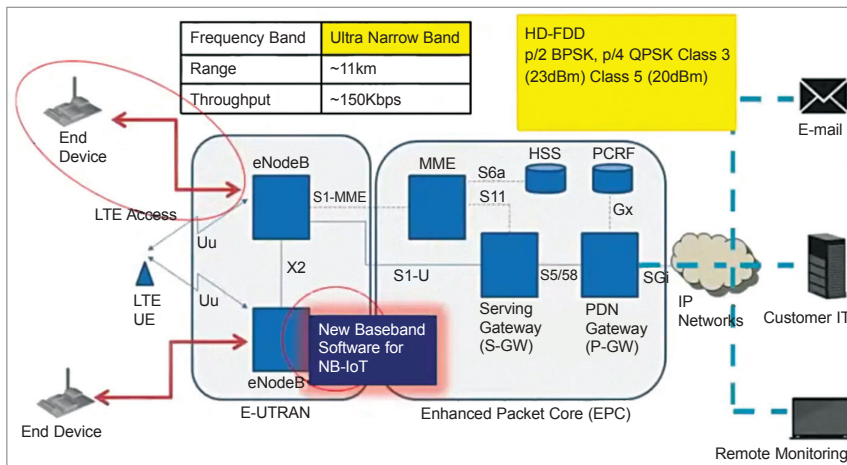


Fig. 6: NB-IoT architecture

3GPP Release	12 (Cat.0) LTE-M	13 (Cat.1.4MHz) LTE-M	13(Cat. 200KHz) NB-IoT
Downlink peak rate	1Mbps	1Mbps	300bps to 200kbps
Uplink peak rate	1Mbps	1Mbps	144kbps
Number of antennas	1	1	1
Duplex mode	Half	Half	Half
UE receive bandwidth	20MHz	1.4MHz	200kHz
UE transmit power (dBm)	23	20	23

Fig. 7: NB-IoT and LTM comparison

complexity of IoT is in one single place here.

Why prioritise wireless?

The key considerations for preferring wireless are:

- Radio range
- Power consumption
- Interference
- Data rate
- Antenna size
- Standard versus proprietary

So, the wireless module is interfaced via SPI or I2C with the microcontroller, which in turn will decide the kind of data to be transmitted.

If we go for the licensed spectrum, then it means to go for a wide area range network. On the connectivity side, there is no requirement for gateways. That is why people prefer GSM protocols and network. But the GSM network is designed for voice communications. In fact, all mobile

network operators (MNOs) do not want to support GSM network from an IoT point of view because once IoT devices increase in number, GSM network and voice communications will get interrupted and suffer. Also, the GSM network is very power-hungry. For IoT, we cannot select any such module. Therefore, NB-IoT is needed.

We can go with the unlicensed band also. But for unlicensed bands, the priority of the data and data interface (methodology of interface) is different. From a security point of view, the unlicensed band needs to have more securities. We need to take care of the hardware and network side. The GSM network topology, however, has already taken care of security.

The end device can work over WAN, GSM, and 2G/3G/4G/5G. And this end device interfaces with the cloud. Therefore, we need to write protocols for the APIs. These protocols have to be decided by following the standards available. The GSM already has 3GPP standards and AT command, which makes it very easy to interface with the device as well as the cloud. However, GSM is power-hungry, and for IoT we require low power. So, LPWAN and NB-IoT are found suitable as they offer low-power data.

Narrowband IoT

Narrowband (NB) standards are based on low power wide area network (LPWAN) technology. It enables a wide range of new IoT devices and services using the GSM platform.

Its features include:

Uses LTE design extensively. These include single carrier frequency division multiple access (SC-FDMA) and frequency division multiple access (FDMA). 3G and 4G are also used on the same spectrum and same bandwidth.

Low cost. Narrowband supports both low-packet and low-power. So, the GSM 2G and 4G networks come together in 3GPP for Narrowband.

Extended coverage. 164dB maximum coupling loss or link budget in comparison to GPRS link budget of 144dB and LTE of 142.7dB.

Low receiver sensitivity. The sensitivity completely depends on the network. As for NB, 2G, and 3G, the sensitivity is high.